
VDD_FE
AGND
VDD_CGEN
DVDD
REF_VDD
VDD_ADC
VDD_DRIVER
VDD10
VSS10
REF_VCM
VDD_DRIVER
AGND
Function: J1 is a single AGND pin position. From V6 it is not fitted (do not populate, not in the BOM, not in the position file); the plated hole and its AGND connection stay on the board.
Source / changes: Inherited from the ASIC designer's board pair, changed in V6. Reference: meeting with the ASIC designer, 12 Nov 2026, notes V1, answer E6 (J1: "Not needed."). file 2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md (a project record, no web address). Changed against V5: only the three part flags: footprint, position, hole and net are as in V5.
chipU01_BCD_OFF_CHIP
U01_BCD1
CLKH
CHIP_RESET
RST_DIGITAL
DGND
DVDD
AGND
REF_VCM
REF_VDD
VDD_ADC
VDD_DRIVER
VDD_FE
VDD10
VSS10
P10_R
M10_R
DISC_OUT
EN_LDO
DATAL<4>
DATAL<3>
DATAL<2>
DATAL<1>
RST_DRIVER
BITL<1>
BITL<2>
BITL<3>
AC_IN
SPI_LDO_P1
DGND1
CGEN_RST
REF_1
REF_2
IMP_TST
RST_FEL
SPI_CLK
SPI_LATCH
SPI_DINL
SPI_DINR
DGND2
SPI_LDO_P2
VDD_CGEN
AGND_1
SPI_LDO_D
FPGA_START
FPGA_CLK
FPGA_EN
FPGA_CHBL
FPGA_CHBR
REF_3
REF_4
RST_FER
BITR<1>
BITR<2>
BITR<3>
SPI_LDO_P
SYNC
CLK32MHZ
DATAR<4>
DATAR<3>
DATAR<2>
DATAR<1>
READ
DGND3
DGND
T_D<1>
T_D<2>
T_D<3>
T_D<4>
CHIP_RESET
AC_IN
DVDD
T_SPI_CLK
T_SPI_LATCH
T_SPI_DINL
T_SPI_DINR
T_SYNC
T_CLK32MHZ
T_D<5>
T_D<6>
T_D<7>
T_D<8>
T_READ
AGND
FE_RESET
IMP_TEST
STIM_START
STIM_CLK
STIM_EN
STIM_CHB
VDD_CGEN
J3
Conn_02x40_Odd_Even
J4
Conn_02x40_Odd_Even
Function: the DISC_OUT finger of U01_BCD1 is left open (no-connect flag). Nothing on this board or on the FPGA side reads it.
Source / changes: Inherited from the ASIC designer's board pair, changed in V6. Reference: meeting with the ASIC designer, 12 Nov 2026, notes V1, answer A3 (DISC_OUT is not needed: "It does not even need to be connected to the chip."). file 2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md (a project record, no web address). Changed against V5: label T_DISC_OUT and its 8.89 mm stub wire removed, no-connect flag added on the pin; no part added or removed. Connector pins 3 and 5 keep the names DISC_OUT(2) and DISC_OUT and still pass J3 to J4. Not verified: this rests on the designer's spoken answer; no chip data sheet page states that the pad may float.
chipU01_BCD_OFF_CHIP
U01_BCD2
Sheet: /
File: ASIC_Top_Board_V7.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1